Highlights

UVHF: Unified Varied-Horizon Forecasting with Multi-Scope Decoder and Balanced Co-Adaptation

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- Unified varied-horizon forecasting is formalized through nested trajectory prefixes.
- A Multi-Scope Decoder adaptively combines forecasts from multiple sharing scopes.
- Balanced Co-Adaptation jointly optimizes scope forecasts and their allocation.
- One model achieves state-of-the-art accuracy across seven forecasting benchmarks.

UVHF: Unified Varied-Horizon Forecasting with Multi-Scope Decoder and Balanced Co-Adaptation

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Abstract

Most time-series forecasting methods optimize a model for a predefined prediction horizon, whereas practical applications often request forecasts of different lengths from the same history. Under the prevailing horizon-specific paradigm, separate models are optimized for different prediction lengths. This fragments multi-horizon service and can produce inconsistent forecasts for future steps shared across horizon requests. We define unified varied-horizon forecasting as learning one horizon-agnostic trajectory whose nested prefixes serve different request endpoints. This formulation imposes two coupled requirements on decoder design. First, predictions for shared future steps must remain invariant to the requested horizon, a property we term cross-horizon prefix consistency (CHPC). Second, jointly modeling short-, medium- and long-range futures requires the decoder to regulate how broadly history-conditioned information is shared across the future domain. We therefore introduce UVHF, a unified forecasting framework built around a Multi-Scope Decoder (MSD). The MSD constructs region-wise forecasts under multiple sharing scopes and integrates them through target-adaptive allocation into one trajectory, thereby satisfying CHPC by construction. We further develop Balanced Co-Adaptation (BCA), which balances the joint optimization of scope-conditioned forecasts and their allocation through scope-wise supervision and usage regularization. Experiments across seven multivariate benchmarks show that UVHF achieves state-of-the-art accuracy against recent horizon-specific and unified forecasters while serving all supported horizons with a single model. It also offers a favorable trade-off between checkpoint storage and inference memory. Controlled ablations support the proposed components, and generalization studies demonstrate the portability of the MSD across the evaluated backbone families. These results establish output-side multi-scope generation as an effective foundation for unified varied-horizon forecasting.

Keywords: Time series forecasting, Unified varied-horizon forecasting, Cross-horizon prefix consistency, Multi-Scope Decoder, Balanced Co-Adaptation, Adaptive forecast generation.

1. Introduction

Multi-horizon forecasts support decisions over multiple planning ranges, from short-term control to long-term scheduling [1]. Yet most long-term time-series forecasting models and benchmark protocols remain horizon-specific: a separate model is trained for each forecast horizon H, such as 96, 192, 336, and 720 steps [2, 3]. Recent work, including ElasTST[4] and time-series foundation models such as TimesFM[5] and Time-MoE[6], has begun to support varied or flexible forecasting horizons through autoregressive and universal-generation designs [7, 8, 9]. However, these efforts remain sparse relative to the extensive horizon-specific literature and have not yet established varied-horizon forecasting as an explicit task with systematic problem analysis and targeted decoder design. In this work, we investigate how the decoder should organize output-side representations across different parts of the future domain.

The horizon-specific protocol fragments forecasting into independent predictors. As illustrated in Fig. 1a, models trained for different horizons can assign different values to the same future time step, despite identical history and

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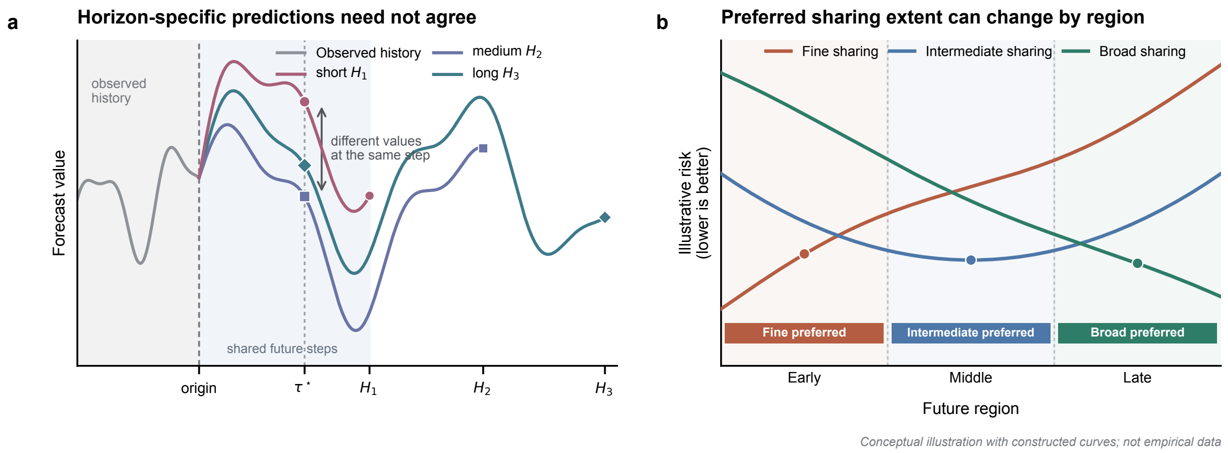

Figure 1: Two challenges in unified varied-horizon forecasting. **a**, Horizon-specific predictors may disagree at the same future time step $\tau^\star$ despite identical observed history. **b**, The sharing extent associated with the lowest risk can vary across future regions.

forecast origin. Their outputs can therefore disagree on overlapping future targets rather than form nested views of a common trajectory. Supporting several horizons further entails separate training, checkpoint storage, deployment and maintenance.

To formalize a unified alternative, we define **unified varied-horizon forecasting** as the setting in which one model serves different request endpoints through a shared prediction trajectory. The task learns a single horizon-agnostic mapping from observed history and future-step index to prediction. Under this mapping, a future-step prediction depends on the history and its step index, but not on the requested horizon. For any $H_1 < H_2$, predictions over steps $1{:}H_1$ therefore remain identical under both requests. We call this requirement **cross-horizon prefix consistency (CHPC)**.

CHPC specifies how forecasts at different horizons should relate, but not how a decoder should generate different future regions. Most architectural advances focus on history encoding, while the output stage often uses a uniform mechanism for all future steps [10, 11, 12, 13]. Such a decoder fixes the sharing extent, that is, how broadly a history-conditioned latent state is reused before step-specific generation. Broad sharing can capture persistent trajectory structure but may smooth local variations. Finer sharing offers greater step-specific flexibility but weaker structural regularization. The preferred balance can vary across samples, variables and future regions, making a fixed extent inadequate for the full forecast domain. Fig. 1b summarizes this output-side heterogeneity, which we term **future-region sharing-demand heterogeneity**. We examine this problem in greater detail in Section 3.

Motivated by this heterogeneity, we introduce **UVHF**, a unified forecasting framework comprising a **Multi-Scope Decoder (MSD)** and **Balanced Co-Adaptation (BCA)**. The MSD assigns each scope a dedicated history projection within one scope-indexed forecast field. Each scope specifies the number of future steps over which a history-conditioned latent state is reused. Target-adaptive allocation then integrates the scope-conditioned forecasts for each sample, variable and future step. The resulting trajectory can adapt its sharing composition across future regions while retaining a single horizon-agnostic prediction function. BCA supports joint learning by supplying direct prediction signals to every scope and regulating aggregate scope usage during optimization. It operates only during training, adds no inference parameters or paths, and preserves the CHPC-compliant forward computation of UVHF.

The contributions of our work are summarized as follows:

1. We formulate unified varied-horizon forecasting as a forecasting system in which CHPC is a basic requirement, and identify future-region sharing-demand heterogeneity as an output-side challenge.

2. We introduce UVHF and its Multi-Scope Decoder, which integrates forecasts generated under multiple sharing scopes through target-adaptive allocation.

3. We develop Balanced Co-Adaptation to support balanced scope learning without increasing inference-time complexity.

Experiments across datasets from multiple application domains show that UVHF achieves state-of-the-art performance among the evaluated baselines and outperforms separately trained horizon-specific forecasters. Component-wise ablations support the utility of the proposed components, and generalization studies demonstrate the portability of the MSD across the evaluated backbone families.

2. Related Work

2.1. Fixed-horizon multi-step forecasting

Multi-step forecasting has traditionally assumed a fixed forecast horizon and differs in how predictions within that horizon are generated. Recursive forecasting iterates one one-step model and feeds predictions back as inputs, which can accumulate errors [14]. Direct forecasting instead trains a separate model for each future step, avoiding recursive propagation but increasing estimation variance and weakening dependencies among future outputs [14]. The multiple-input multiple-output (MIMO) strategy predicts the complete future vector with one multi-output model. The direct–MIMO (DIRMO) strategy instead partitions the horizon into blocks and assigns one multi-output model to each block [15, 16, 17]. These strategies primarily trade error propagation, estimation variance, output dependence and computational cost within a prespecified horizon.

Modern long-term forecasting benchmarks largely use fixed-window, MIMO-like prediction. DLinear, PatchTST and iTransformer map a history representation to one configured prediction length; standard evaluations train separate models for horizons 96, 192, 336 and 720 [1, 2, 3]. This protocol optimizes each horizon independently but does not require agreement on overlapping future targets. Unified varied-horizon forecasting instead uses one model to serve multiple request endpoints and returns prefix-consistent views of one predicted trajectory.

2.2. Unified varied-horizon forecasting

Advances in time-series foundation models have encouraged models that operate across forecasting tasks or output lengths. TimesFM uses decoder-only pretraining across datasets, granularities and horizons, while Timer and Time-MoE generate flexible forecast lengths autoregressively [5, 7, 6]. Chronos and Moirai further extend pretrained forecasting through tokenized probabilistic generation and universal masked-encoder training, respectively [8, 9]. UniTS accommodates heterogeneous time-series tasks through task tokenization [18]. Although these models support multiple horizons or task specifications, they primarily pursue universal pretraining, task unification or scalable generation without systematically treating unified varied-horizon forecasting as a distinct task. ElasTST is a notable exception. It combines structured attention masks, tunable RoPE, multi-scale history patches and horizon reweighting to improve robustness across inference horizons [4]. Its multi-scale mechanism, however, varies history-patch resolution rather than the output-side extent of state sharing. Our work targets this complementary problem by formalizing CHPC and modeling how state-sharing demand changes across future regions.

2.3. Forecast generation and output-side modeling

Most recent forecasting architectures concentrate on extracting and representing information from the observed history, for example through decomposition, frequency enhancement, cross-dimension attention or convolutional token modeling [11, 12, 13, 19, 20]. Their output stages often remain shallow fixed-window projections. DLinear directly maps temporal inputs to future steps, while PatchTST and iTransformer project patch-level or variate-level representations through linear prediction heads [1, 2, 3]. Consequently, the forecasting phase is commonly treated as a readout of an increasingly sophisticated history Encoder rather than a separate modeling problem.

Beyond shallow output projections, several studies have introduced structured mechanisms for forecast generation. TFFS combines common and step-specific future features, N-BEATS and N-HiTS synthesize forecasts through basis expansion or hierarchical interpolation, and Implicit Forecaster composes predicted waves into a global trajectory [21, 22, 23, 24]. PDT learns a prediction-oriented latent domain and combines linear extrapolation of dominant components with an encoder route for complex dynamics [25]. These approaches structure predictive representations or forecast components. The MSD in UVHF explicitly models how broadly a history-conditioned state is shared across future steps and constructs region-local forecasts under multiple scopes before step-specific generation.

2.4. *Multi-scale forecasting and adaptive allocation*

Most multi-scale forecasters extract and combine temporal dependencies from the observed history to form hierarchical representations. SCINet recursively interacts with downsampled subsequences, while N-HiTS couples multi-rate sampling with hierarchical synthesis [26, 23]. Pathformer selects pathways over input patch scales, whereas TimeMixer mixes decomposed histories with scale-specific predictors [27, 28]. TCDNet models intra-scale dependencies during temporal compression and integrates time–frequency features into decompression to the forecast horizon [29]. These methods organize scales through temporal resolution, frequency structure or prediction branches. The MSD defines scale through output-side state reuse. Starting from one History State, it generates forecasts under multiple state-sharing scopes and allocates these granularities according to each future target.

Expert-based forecasters also use conditional allocation, but route among components with different modeling functions. MoLE mixes complete linear-centric forecasters, whereas FreqMoE assigns frequency components to experts [30, 31]. Time-MoE and Moirai-MoE instead activate sparse internal experts within foundation models [6, 32]. The scopes in the MSD are jointly learned forecast-generation paths with shared Encoder and Region-to-Step Forecast Generator parameters. Scope Projections construct information pools tied to explicit state-reuse extents, and Scope Probabilities allocate these sharing granularities for each target. BCA jointly optimizes the resulting scope-indexed forecast field and its allocation process. This output-side interpretation underlies the sharing-demand heterogeneity formalized in Section 3.

3. Problem Formulation and Empirical Motivation

The preceding discussion shows that unified varied-horizon forecasting involves more than sharing parameters across prediction lengths. A unified forecaster should return coherent predictions for future steps shared by different horizon requests, while its decoder must determine how predictive information is shared across the future domain. We now formalize these two issues and examine them through controlled empirical analyses, which motivate the decoder design developed in Section 4.

3.1. *Unified varied-horizon forecasting and cross-horizon prefix consistency*

Consider two forecasting requests with horizons $H_i < H_j$ issued from the same observed history. Their first H_i steps refer to identical future targets and should therefore receive identical predictions. We formalize this requirement below.

At forecast origin o, let

$$\mathbf{X}_o = [\mathbf{x}_{o-L+1}, \ldots, \mathbf{x}_o] \in \mathbb{R}^{L\times C}, \qquad \mathbf{Y}_o^{(H)} = [y_{o+\tau,c}]_{\tau=1,\ldots,H;\ c=1,\ldots,C} \in \mathbb{R}^{H\times C}. \tag{1}$$

Here, $\mathbf{X}_o$ is a length-L multivariate history, and $\mathbf{Y}_o^{(H)}$ contains the next H targets for C variables. Let $\mathcal{H} = \{H_1, \ldots, H_M\}$ denote the supported horizons and $T = \max \mathcal{H}$. The horizon H specifies the request endpoint, whereas τ indexes a particular future step. Thus, target (τ, c) is shared by all requests with $H \geq \tau$.

This shared-target relation defines **cross-horizon prefix consistency (CHPC)**. Let $\widehat{y}_{o+\tau,c}^{(H)}$ denote the prediction returned for target (τ, c) under a request with horizon H. Given identical history, forecast origin and preprocessing, CHPC requires

$$\widehat{y}_{o+\tau,c}^{(H_i)} = \widehat{y}_{o+\tau,c}^{(H_j)}, \qquad \forall\, H_i, H_j \in \mathcal{H},\ H_i < H_j, \quad 1 \leq \tau \leq H_i, \quad 1 \leq c \leq C. \tag{2}$$

Conventional long-term forecasting learns an independent predictor f_{θ_H} for each horizon. Because both parameters and optimization are horizon-specific, these predictors are not constrained to satisfy CHPC on their overlapping outputs.

A varied-horizon forecaster should instead use a single **future-step-indexed prediction function** $g_\theta(\mathbf{X}_o, \tau, c)$ and construct an H-step forecast as

$$\widehat{\mathbf{Y}}_o^{(H)} = [g_\theta(\mathbf{X}_o, \tau, c)]_{\tau=1,\ldots,H;\ c=1,\ldots,C}\,. \tag{3}$$

For a varied-horizon forecaster, all horizon requests query the same function at a shared target, so CHPC holds by construction. Changing the requested horizon alters only the returned prefix length, and each request becomes a nested view of one predicted trajectory.

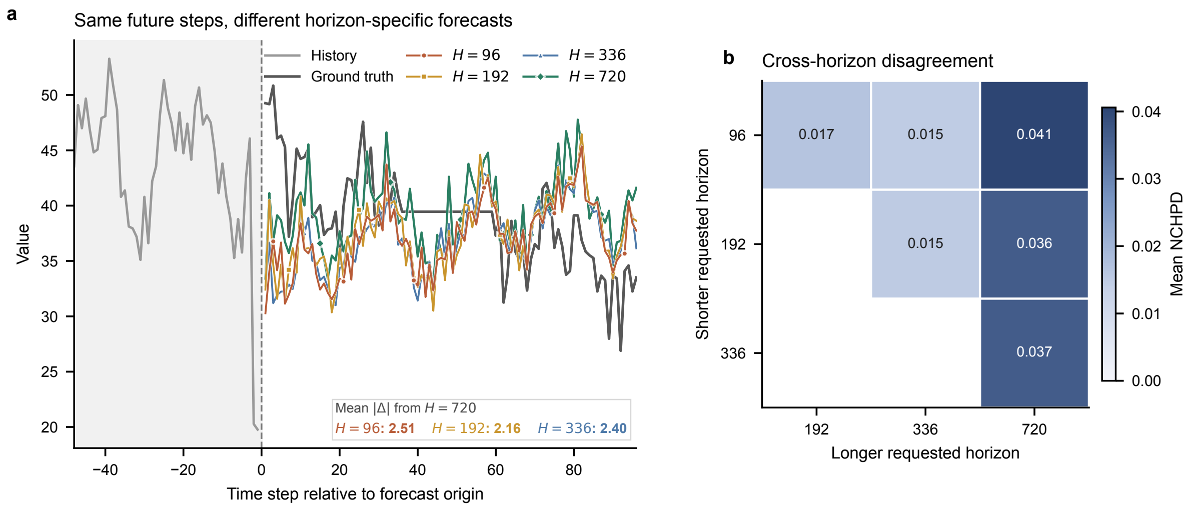

Figure 2: Horizon-specific forecasts disagree on shared future steps. **a**, Selected ETTh2 validation trajectory from independently trained DLinear models for horizons 96, 192, 336 and 720; the inset reports differences from the $H = 720$ forecast. **b**, NCHPD over 2,161 aligned ETTh2 validation origins and all variables.

3.2. Horizon-specific prefix inconsistency

Independently trained horizon-specific predictors may agree on some overlapping outputs, but their objectives do not enforce CHPC. We refer to a violation of CHPC as **cross-horizon prefix inconsistency** and quantify its magnitude through prediction disagreement on aligned histories and shared future targets.

At origin o, let $\widehat{\mathbf{Y}}_o^{(H_i)} \in \mathbb{R}^{H_i \times C}$ and $\widehat{\mathbf{Y}}_o^{(H_j)} \in \mathbb{R}^{H_j \times C}$ denote forecasts from independently trained models, where $H_i < H_j$. We define their **cross-horizon prefix disagreement (CHPD)** as

$$\mathrm{CHPD}_o(H_i, H_j) = \frac{1}{H_i C} \sum_{\tau=1}^{H_i} \sum_{c=1}^{C} \left| \widehat{y}_{o+\tau,c}^{(H_i)} - \widehat{y}_{o+\tau,c}^{(H_j)} \right|. \tag{4}$$

CHPD measures the mean absolute difference over the shared prefix in the original data scale. To aggregate variables with different magnitudes, we further define normalized CHPD (NCHPD):

$$\mathrm{NCHPD}(H_i, H_j) = \frac{1}{|O|C} \sum_{o \in O} \sum_{c=1}^{C} \frac{\frac{1}{H_i} \sum_{\tau=1}^{H_i} \left| \widehat{y}_{o+\tau,c}^{(H_i)} - \widehat{y}_{o+\tau,c}^{(H_j)} \right|}{\sigma_c^{\mathrm{train}} + \epsilon}. \tag{5}$$

Here, O denotes the aligned evaluation origins, $\sigma_c^{\mathrm{train}}$ is the train-split standard deviation of variable c, and ϵ is the fixed numerical stabilizer used in the computation. All comparisons use identical histories, forecast origins, scalers and overlapping target indices.

To visualize this inconsistency, we compare multi-horizon forecasts from DLinear models independently optimized on ETTh2 [1]. Fig. 2a shows that their predictions diverge over the same 96 future steps: relative to the $H = 720$ forecast, the forecasts for $H \in \{96, 192, 336\}$ differ by 2.51, 2.16 and 2.40 in mean absolute raw scale, respectively.

We further evaluate all 2,161 aligned ETTh2 validation origins and variables, with the resulting NCHPD matrix shown in Fig. 2b. NCHPD is non-zero for every horizon pair and is more pronounced between the longest and shorter requested horizons: the $H = 96$, $H = 192$ and $H = 336$ comparisons with $H = 720$ yield 0.0406, 0.0365 and 0.0366, whereas pairs among the three shorter horizons range from 0.0148 to 0.0166. These results show that the independently optimized DLinear models evaluated here fail to form a prefix-consistent prediction trajectory, reflecting a structural limitation of horizon-specific forecasting: independently optimized models are not constrained to agree on shared future targets.

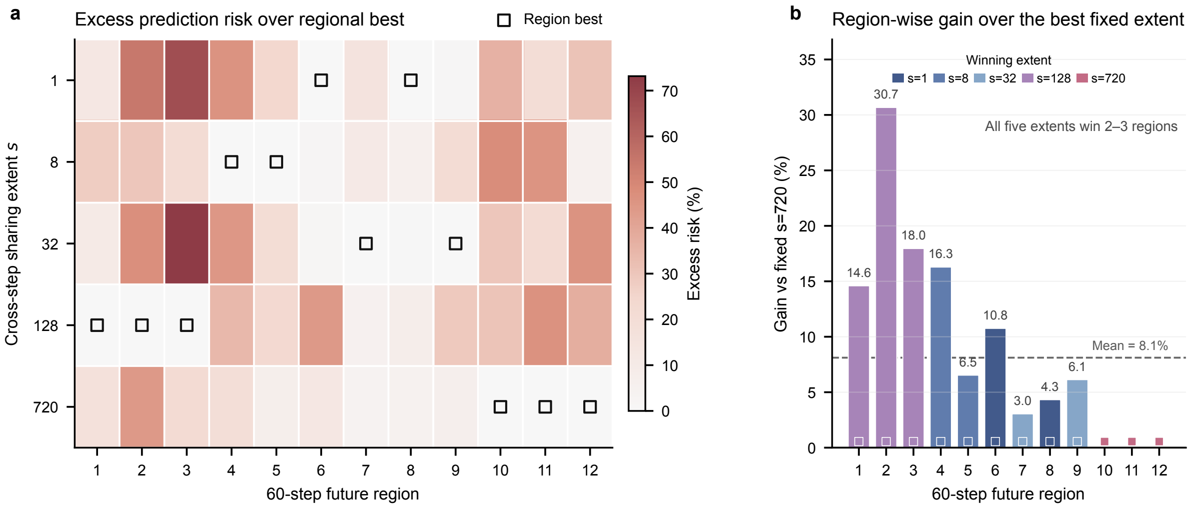

Figure 3: Sharing preferences vary across future regions. **a**, Percentage excess prediction risk of five capacity-matched single-extent predictors on a selected ETTm2 validation example; squares mark regional winners. **b**, MSE reduction from regional selection relative to the fixed extent $s = 720$; dashed line, mean over 12 regions.

3.3. Future-region sharing-demand heterogeneity

CHPC specifies how predictions from different horizon requests should relate, but it does not determine how a unified decoder should construct the shared future trajectory. Simply adapting an architecture designed for one fixed horizon leaves this output-side requirement unresolved: a varied-horizon forecaster must learn from requests spanning short to long endpoints while producing one prefix-consistent trajectory. The decoder is therefore central to how predictive information is organized across the future domain, particularly in determining how broadly each history-conditioned state is reused before step-specific prediction.

We use the **sharing extent** s to denote the number of future steps that reuse one latent state. A **future region** $\mathcal{B}_b \subseteq \{1, \ldots, T\}$ is a contiguous subset of the maximum future domain rather than a requested horizon. Broad sharing can capture persistent trajectory structure but may smooth local variations, whereas fine sharing offers greater local flexibility with weaker cross-step regularization. We refer to variation in the preferred extent across regions as **future-region sharing-demand heterogeneity**.

To isolate this factor, we construct capacity-matched single-extent predictors that differ only in sharing extent; all other architecture, training and evaluation settings remain identical.

For aligned origin o, we define the **future-region prediction risk** of sharing extent s over region $\mathcal{B}_b$ as the empirical squared-error loss

$$R_{o,b,s} = \frac{1}{|\mathcal{B}_b| C} \sum_{\tau \in \mathcal{B}_b} \sum_{c=1}^{C} \left(\widehat{y}^{(s)}_{o+\tau,c} - y_{o+\tau,c} \right)^2 . \tag{6}$$

For a given origin and region, $R_{o,b,s}$ averages squared errors over the region's future steps and variables. Lower $R_{o,b,s}$ indicates that extent s better matches region $\mathcal{B}_b$ within the controlled family; a region-dependent preference appears when $s^{\star}_{o,b} = \arg\min_s R_{o,b,s}$ changes with b.

We evaluate the five predictors with $s \in \{1, 8, 32, 128, 720\}$ across 12 contiguous 60-step future regions on ETTm2. Fig. 3a reports the percentage excess prediction risk of each extent above the lowest-risk extent within each region, with outlined squares marking the region-wise winners. The preferred extents vary across the future domain: each of the five extents wins two or three regions, and the mean best-versus-second-best margin reaches 10.266%. No fixed extent minimizes prediction risk throughout this controlled example, supporting heterogeneous sharing demand across future regions.

Fig. 3b quantifies the performance upper bound associated with these region-wise preferences. Relative to the best fixed extent ($s = 720$), selecting the lowest-risk extent separately for each region reduces average MSE by

8.112%. Because the regional winners are selected using validation labels across separately trained predictors, this value represents descriptive oracle headroom rather than the realized gain of a learned decoder. Within this controlled example, its magnitude nevertheless shows that one fixed extent can leave meaningful region-specific headroom, motivating a decoder that can adapt sharing across the future domain.

4. UVHF Framework

Section 3 established two decoder-side requirements for unified varied-horizon forecasting. Predictions for a shared future target should be invariant to the requested horizon, while the decoder should not impose one latent-state sharing extent on the entire future domain. UVHF addresses these requirements through two coordinated components. Its **Multi-Scope Decoder (MSD)** constructs one future-step-indexed trajectory from multiple sharing granularities, and **Balanced Co-Adaptation (BCA)** jointly optimizes the scope-indexed forecast field and its allocation process.

4.1. Architecture overview

Fig. 4 summarizes the forward computation of the MSD, which comprises a **Scope Forecasting Path** and a **Target-Adaptive Allocation Path**. The decoder consumes a variable-wise History State rather than relying on a particular Encoder. Any Encoder that models temporal patch tokens and returns the required tensor interface can therefore serve as its backbone. This interface covers the patch-based Encoder family commonly used in time-series forecasting [2, 19]. Given a History Series $\mathbf{X} \in \mathbb{R}^{B\times L\times C}$, Patchify and the Encoder produce $\mathbf{R} \in \mathbb{R}^{B\times C\times R}$. The Scope Forecasting Path constructs region-wise predictions under each sharing scope and assembles them into $\mathcal{F}_\theta(\mathbf{X}) \in \mathbb{R}^{B\times C\times T\times S}$. The Target-Adaptive Allocation Path then determines how each future target draws on these sharing granularities.

The Target-Adaptive Allocation Path combines a projected History State with the Future Coordinate $\boldsymbol{\phi}_\tau$ and produces Scope Probabilities $\boldsymbol{\Pi} \in \mathbb{R}^{B\times C\times T\times S}$. Weighted contraction with the scope-indexed forecast field yields one trajectory $\widehat{\mathbf{Y}} \in \mathbb{R}^{B\times T\times C}$. For an H-step request, the MSD evaluates only the region–target pairs intersecting the first H future steps. Shared targets retain the same region-local computation across request endpoints, so varying H changes the evaluated extent while preserving all overlapping predictions. UVHF thereby supports variable-length outputs and satisfies CHPC by construction.

The Scope-conditioned Forecasts and Scope Probabilities are optimized jointly. BCA separates two training roles: the Uniform-Prefix Forecasting Loss optimizes varied-horizon prediction, while the Scope-Wise Forecasting Loss and Allocation-Balance Regularizer stabilize multi-scope learning.

4.2. History state and future coordinate

The MSD begins with the History State and Future Coordinate shown on the left of Fig. 4. The History Series is normalized per variable, divided into P patches by Patchify and processed by the shared Encoder. This produces $\mathbf{Z} \in \mathbb{R}^{B\times C\times P\times D_e}$, which is flattened over its patch and embedding dimensions:

$$\mathbf{Z} = \mathcal{E}(\mathcal{N}(\mathbf{X})), \qquad \mathbf{r}_{b,c} = \mathrm{vec}(\mathbf{Z}_{b,c}) \in \mathbb{R}^R, \qquad R = PD_e. \tag{7}$$

Collecting $\mathbf{r}_{b,c}$ over samples and variables gives the History State $\mathbf{R} = [\mathbf{r}_{b,c}] \in \mathbb{R}^{B\times C\times R}$. The preserved variable axis provides one history representation for each sample-variable pair, which is used by both decoder paths.

The History State summarizes the observed series, but a unified decoder must also identify where each prediction lies in the future domain. Using the requested horizon for this purpose would assign different conditioning contexts to the same target under different requests. The MSD therefore introduces a **Future Coordinate** for every future step. This fixed coordinate supplies a horizon-independent target identity and a common positional basis for regions at different scopes. It also allows the Target-Adaptive Allocation Path to vary its scope preference across future steps.

Formally, the Future Coordinate is the field $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_T]^\top \in \mathbb{R}^{T\times D_q}$. We construct it from low-order discrete cosine functions, which provide smooth coordinate channels at progressively finer temporal frequencies. For $d = 0, \ldots, D_q - 1$, we first define

$$\widetilde{\phi}_{\tau,d} = \cos\left(\frac{\pi(\tau - \frac{1}{2})d}{T}\right). \tag{8}$$

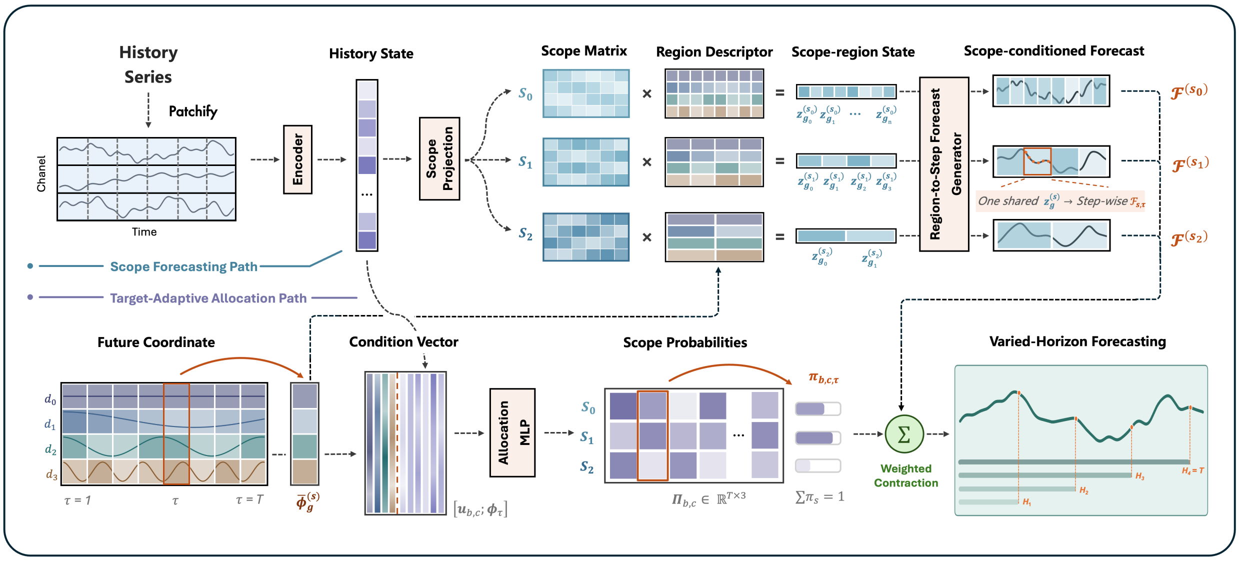

Figure 4: The Multi-Scope Decoder in UVHF. Scope-conditioned forecasts are generated region-wise and integrated through target-adaptive allocation into a prefix-consistent trajectory.

The constant coordinate is $\phi_{\tau,0} = 1$. Each nonconstant coordinate is centered over the future domain and scaled as

$$\phi_{\tau,d} = \sqrt{2}\left(\widetilde{\phi}_{\tau,d} - \frac{1}{T}\sum_{j=1}^{T}\widetilde{\phi}_{j,d}\right), \qquad d \geq 1. \tag{9}$$

The constant channel preserves a global reference, while the centered nonconstant channels distinguish positions across the future domain. The Future Coordinate field serves two roles. Region-wise averaging produces the Region Descriptors used to generate Scope-conditioned Forecasts, whereas the unpooled coordinate $\boldsymbol{\phi}_\tau$ identifies the individual target in the Condition Vector used for Scope Probabilities.

4.3. *Generation of scope-conditioned forecasts*

The Scope Forecasting Path contains S parallel scope lines, one for each sharing scope in $\mathcal{S} = \{s_1, \ldots, s_S\}$. Each scope has its own parameterized Scope Projection, which maps the History State into a dedicated Scope Matrix:

$$\mathbf{M}_{b,c}^{(s)} = \mathrm{reshape}_{D_q\times K}\left(\mathbf{r}_{b,c}\mathbf{W}^{(s)} + \mathbf{b}^{(s)}\right) \in \mathbb{R}^{D_q\times K}. \tag{10}$$

The matrix form preserves a Future-Coordinate axis and a latent-mode axis, allowing Region Descriptors to query scope-specific history information at different future locations without assigning a separate prediction head to every region.

For a scope s, the future domain is divided into contiguous regions

$$\mathcal{G}_g^{(s)} = \{(g-1)s + 1, \ldots, gs\}, \qquad g = 1, \ldots, T/s. \tag{11}$$

The Future Coordinate within each region is averaged to form the Region Descriptor, which provides a compact positional summary for region-wise state construction:

$$\overline{\boldsymbol{\phi}}_g^{(s)} = \frac{1}{s}\sum_{\tau\in\mathcal{G}_g^{(s)}} \boldsymbol{\phi}_\tau \in \mathbb{R}^{D_q}. \tag{12}$$

The Region Descriptor selects and combines information from this matrix through a coordinate contraction, producing the region-wise Scope-region State

$$\mathbf{z}^{(s)}_{b,c,g} = \left(\overline{\boldsymbol{\phi}}^{(s)}_{g}\right)^{\top} \mathbf{M}^{(s)}_{b,c} \in \mathbb{R}^{K}. \tag{13}$$

Once the Scope Matrix is available, each Scope-region State depends only on the descriptor of its own region. Regions can therefore be evaluated separately and in parallel, although they draw on the same scope-specific history information. All future steps in $\mathcal{G}^{(s)}_{g}$ reuse $\mathbf{z}^{(s)}_{b,c,g}$. A finer scope constructs more region states and limits reuse to shorter intervals, whereas a broader scope shares each state over a longer interval. Scope therefore defines the cross-step reuse pattern.

The Region-to-Step Forecast Generator converts each shared Scope-region State into step-wise predictions. Let $g_s(\tau)$ denote the region containing step τ. The generator is shared across scopes and regions, and uses step-specific vectors $\mathbf{a}_\tau, \mathbf{n}_\tau \in \mathbb{R}^{K}$ and bias β_τ to define

$$\mathcal{F}_{b,c,\tau,s} = \left\langle \mathbf{a}_\tau, \mathbf{z}^{(s)}_{b,c,g_s(\tau)} \right\rangle + \left\langle \mathbf{n}_\tau, \mathrm{GELU}\left(\mathbf{z}^{(s)}_{b,c,g_s(\tau)}\right) \right\rangle + \beta_\tau. \tag{14}$$

Concatenating the separately generated regions gives the Scope-conditioned Forecast $\mathcal{F}^{(s)}$ for scope s. Collecting these forecasts produces the scope-indexed field $\mathcal{F}_\theta(\mathbf{X}) \in \mathbb{R}^{B\times C\times T\times S}$. The MSD constructs this field as a unified decoder through scope-specific projections and a shared Encoder and Region-to-Step Forecast Generator. Parameter sharing couples forecast generation across scopes while avoiding duplicated Encoder computation and generator parameters.

4.4. Target-adaptive scope allocation

Different future targets may require information organized at different sharing granularities. Estimating this preference requires both the dynamics encoded in the observed history and the target position in the future domain. The Target-Adaptive Allocation Path represents these two signals using a compact history summary and the Future Coordinate. It first projects the History State as

$$\mathbf{u}_{b,c} = \mathbf{W}_h \mathbf{r}_{b,c} + \mathbf{b}_h \in \mathbb{R}^{D_h}. \tag{15}$$

The compact summary $\mathbf{u}_{b,c}$ retains the sample-variable context used for allocation. For future step τ, the Condition Vector concatenates this summary with $\boldsymbol{\phi}_\tau$. The Allocation MLP maps the resulting vector to scope logits

$$\boldsymbol{\ell}_{b,c,\tau} = \mathbf{W}_o\,\mathrm{GELU}\left(\mathbf{W}_p[\mathbf{u}_{b,c}; \boldsymbol{\phi}_\tau] + \mathbf{b}_p\right) + \mathbf{b}_o \in \mathbb{R}^{S}, \tag{16}$$

$$\pi_{b,c,\tau,s} = \frac{\exp(\ell_{b,c,\tau,s})}{\sum_{s'=1}^{S} \exp(\ell_{b,c,\tau,s'})}. \tag{17}$$

Softmax converts these logits into the Scope Probability $\pi_{b,c,\tau,s}$ assigned to scope s. Collecting all probabilities gives $\mathbf{\Pi} \in \mathbb{R}^{B\times C\times T\times S}$, with $\sum_{s=1}^{S} \pi_{b,c,\tau,s} = 1$. Each probability vector parameterizes how strongly a future target draws on information organized at different sharing granularities. The final normalized prediction reads from the unified scope-indexed forecast field through weighted contraction:

$$\widehat{y}_{b,\tau,c} = \sum_{s=1}^{S} \pi_{b,c,\tau,s} \mathcal{F}_{b,c,\tau,s}. \tag{18}$$

This contraction is not an average over separately trained model outputs. The Scope Forecasting Path and Target-Adaptive Allocation Path are jointly learned within one decoder, allowing each target to receive a history-conditioned mixture of sharing granularities.

For an H-step request, the MSD needs only the Scope-region States whose regions intersect $1{:}H$ and the allocation entries for $\tau \le H$. The resulting normalized prefix is transformed back to the original data scale and returned as

$$\widehat{\mathbf{Y}}^{(H)}_{b} = [\widehat{y}_{b,\tau,c}]_{\tau=1,\ldots,H;\; c=1,\ldots,C}\,. \tag{19}$$

Changing H restricts the active region and target computations but does not alter any prediction shared with a longer request. Every request is therefore a nested view of the same trajectory. The learned probabilities encode target-adaptive information allocation across scopes; their scope preferences are evaluated empirically in later ablations and internal-behavior analyses.

4.5. Balanced Co-Adaptation

Training the MSD within UVHF introduces two coupled requirements. The fused trajectory requires uniform supervision across supported prefix lengths, while all scope lines must develop reliable forecasts as allocation is learned. BCA therefore combines the **Uniform-Prefix Forecasting Loss** for varied-horizon prediction with two terms that stabilize multi-scope learning: the **Scope-Wise Forecasting Loss** and **Allocation-Balance Regularizer**.

Varied-horizon training should optimize every supported prediction length without privileging one endpoint. We therefore average raw-scale MAE equally over all prefixes $h = 1, \ldots, T$ to define the Uniform-Prefix Forecasting Loss. Before loss evaluation, inverse normalization maps the fused and Scope-conditioned Forecasts to the original data scale. Let $y_{b,\tau,c}$ denote the corresponding raw-scale target. The fused objective is

$$\mathcal{L}_{\text{prefix}} = \frac{1}{T}\sum_{h=1}^{T}\frac{1}{BCh}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{\tau=1}^{h}\left|\widehat{y}_{b,\tau,c} - y_{b,\tau,c}\right|. \tag{20}$$

Exchanging the prefix and future-step summations gives an equivalent dense-prefix measure

$$\omega_\tau = \frac{1}{T}\sum_{h=\tau}^{T}\frac{1}{h}, \qquad \sum_{\tau=1}^{T}\omega_\tau = 1. \tag{21}$$

Under this measure, every prefix endpoint contributes equally, while a future step is weighted by the number of prefixes containing it. The same loss can therefore be written as

$$\mathcal{L}_{\text{prefix}} = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{\tau=1}^{T}\omega_\tau\left|\widehat{y}_{b,\tau,c} - y_{b,\tau,c}\right|. \tag{22}$$

The Uniform-Prefix Forecasting Loss establishes the varied-horizon objective, but it does not resolve the joint optimization of multiple scope lines. Weighted contraction makes the fused-loss gradient received by each scope depend on its current Scope Probability. For a fixed target, this relation gives

$$\frac{\partial\mathcal{L}_{\text{prefix}}}{\partial\mathcal{F}_{b,c,\tau,s}} = \pi_{b,c,\tau,s}\frac{\partial\mathcal{L}_{\text{prefix}}}{\partial\widehat{y}_{b,\tau,c}}. \tag{23}$$

Consequently, early probability concentration can suppress learning in low-probability scope lines before their forecasts mature. The allocator then compares unevenly developed predictions, which can reinforce its initial preference. We first address this imbalance with the Scope-Wise Forecasting Loss, which applies the same prefix measure directly to every Scope-conditioned Forecast:

$$\mathcal{L}_{\text{scope}} = \frac{1}{BCS}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{s=1}^{S}\sum_{\tau=1}^{T}\omega_\tau\left|\mathcal{F}_{b,c,\tau,s} - y_{b,\tau,c}\right|. \tag{24}$$

This term gives every scope a direct prediction gradient independent of its current Scope Probability. Direct supervision alone, however, does not regulate how the fused predictor distributes probability-mediated gradients. We therefore introduce the Allocation-Balance Regularizer using the uniform reference $q_s = 1/S$. We first compute the marginal scope usage over the batch and future domain:

$$\bar{\pi}_s = \frac{1}{BC}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{\tau=1}^{T}\omega_\tau\pi_{b,c,\tau,s}, \qquad s = 1, \ldots, S. \tag{25}$$

The Allocation-Balance Regularizer then constrains this aggregate usage toward the uniform reference:

$$\mathcal{L}_{\text{balance}} = \frac{D_{\text{KL}}(\mathbf{q} \,\|\, \bar{\boldsymbol{\pi}})}{\log S}. \tag{26}$$

The complete objective is

$$\mathcal{L}_{\text{BCA}} = \mathcal{L}_{\text{prefix}} + \lambda_{\text{scope}} \mathcal{L}_{\text{scope}} + \lambda_{\text{balance}}(u) \mathcal{L}_{\text{balance}}, \tag{27}$$

where $u \in [0, 1]$ is optimizer progress. The coefficient $\lambda^{\max}_{\text{balance}}$ specifies the maximum regularization strength, while u_{ramp} denotes the optimizer progress at which the ramp reaches this value. In the frozen configuration, $\lambda_{\text{scope}} = 1$ and

$$\lambda_{\text{balance}}(u) = \lambda^{\max}_{\text{balance}} \min\left(\frac{u}{u_{\text{ramp}}}, 1\right). \tag{28}$$

The regularizer penalizes systematic imbalance in marginal scope usage across the batch and future domain. Because it acts on $\bar{\boldsymbol{\pi}}$, individual future steps and regions can retain distinct, non-uniform scope preferences while the aggregate allocation remains distributed across scopes. Its ramp introduces this pressure gradually as the forecasting paths develop, reducing persistent scope underuse while preserving local allocation patterns. Together with direct scope supervision, it supports more balanced multi-scope optimization.

5. Experiments

UVHF is designed to replace separately optimized horizon-specific predictors with one prefix-consistent forecaster. The central empirical question is whether a single model can serve all requested horizons without sacrificing forecasting accuracy. We examine this question through benchmark comparisons and deployment costs, and then use controlled ablations and validation diagnostics to identify the contribution and behavior of the proposed components.

5.1. Experimental setup

Datasets and metrics. We evaluate UVHF on seven public multivariate forecasting benchmarks: ETTm1, ETTm2, ETTh1, ETTh2, Weather, Electricity (ECL) and Solar. These benchmarks span electricity transformers, meteorology, electricity consumption and solar generation across different sampling frequencies and variable dimensions. Following the standard long-term forecasting protocol, we use prediction horizons $\mathcal{H} = \{96, 192, 336, 720\}$ and report mean squared error (MSE) and mean absolute error (MAE), where lower values indicate better forecasts. Further metric definitions, dataset descriptions, statistics, data splits and preprocessing details are provided in Appendix A.

Model and baselines. UVHF pairs the MSD with a lightweight patch-token MLP Encoder to isolate the contribution of decoder-side multi-scope generation. The comparison covers four representative design families: (1) Transformer-based models, including SimpleTM, iTransformer and PatchTST [33, 3, 2]; (2) lightweight linear or attention-based forecasters and training frameworks, including TimeAlign, QDF and DLinear [34, 35, 1]; (3) convolutional models, including TVNet and ModernTCN [36, 20]; and (4) multi-scale or decomposition-based models, including AMD and TimeMixer [37, 28].

Evaluation protocols. The horizon-specific comparison tests whether one UVHF framework per dataset can retain forecasting accuracy against baselines optimized separately for each requested horizon. The complementary one-model-all-horizons comparison places every method under the same unified workflow. Each model is trained for the maximum horizon, and an H-step request is evaluated from its first H outputs using the corresponding official fixed-H test loader. This protocol evaluates how effectively each architecture serves all requested horizons through one model and one shared prediction trajectory.

Implementation details. Local experiments are implemented in Python 3.12.13 and PyTorch 2.9.0 with CUDA 12.8, and run on NVIDIA GeForce RTX 3090 GPUs. UVHF is trained with BCA using AdamW and a cosine learning-rate schedule [38, 39]. All locally reproduced baselines are built from their official codebases, and any source-informed configuration adaptation is documented explicitly.

Table 1: Comparison with horizon-specific baselines (four-horizon mean MSE/MAE). Average denotes the seven-dataset mean. Red bold and blue underline indicate best and second-best values. Best reports MSE/MAE win counts across datasets.

Dataset	UVHF (Ours)	TimeAlign (2026)	QDF (2026)	AMD (2025)	SimpleTM (2025)	TVNet (2025)	iTransformer (2024)	TimeMixer (2024)	ModernTCN (2024)	PatchTST (2023)	DLinear (2023)
ETTm1	**0.329**/**0.363**	<u>0.341</u>/<u>0.368</u>	0.353/0.380	0.351/0.378	0.383/0.396	0.348/0.379	0.362/0.391	0.356/0.380	0.351/0.381	0.353/0.376	0.356/0.378
ETTm2	<u>0.248</u>/**0.304**	**0.245**/**0.304**	0.257/0.315	0.254/0.315	0.280/0.325	0.251/<u>0.311</u>	0.269/0.329	0.257/0.318	0.253/0.314	0.256/0.317	0.259/0.324
ETTh1	**0.390**/**0.415**	0.418/0.431	0.439/0.442	0.412/0.428	0.428/0.433	0.407/0.421	0.439/0.448	0.427/0.441	<u>0.404</u>/<u>0.420</u>	0.419/0.437	0.425/0.439
ETTh2	**0.306**/**0.364**	0.347/0.387	0.353/0.394	0.366/0.406	0.356/0.392	0.324/<u>0.377</u>	0.374/0.406	0.349/0.397	<u>0.322</u>/0.379	0.351/0.395	0.431/0.447
Weather	**0.214**/**0.245**	<u>0.218</u>/<u>0.247</u>	0.237/0.271	0.225/0.265	0.243/0.271	0.221/0.261	0.233/0.271	0.226/0.264	0.224/0.264	0.226/0.264	0.242/0.293
ECL	**0.151**/**0.244**	<u>0.156</u>/<u>0.246</u>	0.167/0.260	0.162/0.257	0.168/0.262	0.165/0.254	0.164/0.261	0.173/0.272	<u>0.156</u>/0.253	0.159/0.253	0.166/0.264
Solar	**0.188**/**0.208**	0.196/<u>0.217</u>	0.208/0.258	0.205/0.248	<u>0.191</u>/0.251	0.228/0.277	0.202/0.248	0.193/0.252	0.228/0.282	0.194/0.245	0.223/0.226
Average	**0.261**/**0.306**	<u>0.274</u>/<u>0.314</u>	0.288/0.331	0.282/0.328	0.293/0.333	0.278/0.326	0.292/0.336	0.283/0.332	0.277/0.327	0.280/0.326	0.300/0.339
Best	6/7	1/1	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0	0/0

Table 2: One-model-all-horizons comparison (four-horizon mean MSE/MAE). Average denotes the seven-dataset mean. Red bold and blue underline indicate best and second-best values. Best reports MSE/MAE win counts across datasets.

Dataset	UVHF (Ours)	TimeAlign	QDF	AMD	SimpleTM	iTransformer	PatchTST	DLinear
ETTm1	**0.329**/**0.363**	<u>0.345</u>/<u>0.371</u>	0.352/0.384	0.354/0.380	0.384/0.399	0.409/0.415	0.354/0.384	0.359/0.381
ETTm2	**0.248**/**0.304**	<u>0.250</u>/<u>0.308</u>	0.257/0.316	0.255/0.316	0.285/0.330	0.294/0.339	0.258/0.316	0.292/0.359
ETTh1	**0.390**/**0.415**	0.420/0.433	0.452/0.453	<u>0.416</u>/<u>0.430</u>	0.434/0.436	0.468/0.458	0.429/0.440	0.474/0.480
ETTh2	**0.306**/**0.364**	0.368/0.399	<u>0.343</u>/<u>0.392</u>	0.370/0.411	0.356/0.393	0.385/0.408	0.352/0.395	0.415/0.439
Weather	**0.214**/**0.245**	<u>0.220</u>/<u>0.250</u>	0.260/0.295	0.226/0.266	0.254/0.280	0.262/0.284	0.231/0.268	0.247/0.301
ECL	**0.151**/**0.244**	<u>0.157</u>/<u>0.246</u>	0.183/0.281	0.163/0.257	0.169/0.265	0.177/0.269	0.161/0.255	0.167/0.265
Solar	**0.188**/**0.208**	0.193/<u>0.219</u>	0.207/0.268	0.204/0.250	<u>0.190</u>/0.247	0.229/0.261	0.194/0.247	0.253/0.313
Average	**0.261**/**0.306**	<u>0.279</u>/<u>0.318</u>	0.293/0.341	0.284/0.330	0.296/0.336	0.318/0.348	0.283/0.329	0.315/0.363
Best	7/7	0/0	0/0	0/0	0/0	0/0	0/0	0/0

5.2. Comparison with horizon-specific forecasters

Table 1 compares one UVHF framework per dataset with baselines optimized separately for each requested horizon and reports the four-horizon mean for each dataset. Complete horizon-wise results are provided in Appendix B. UVHF obtains the best result in 13 of the 14 dataset–metric comparisons and the second-best result in the remaining comparison. Compared with TimeAlign, the strongest aggregate horizon-specific baseline, UVHF reduces the seven-dataset average MSE and MAE by 4.94% and 2.54%, respectively. The advantage persists across datasets with markedly different dimensionalities. Over the four low-dimensional ETT benchmarks, UVHF improves the average MSE and MAE over TimeAlign by 5.69% and 2.89%; on the high-dimensional ECL and Solar datasets, the corresponding improvements are 3.90% and 2.27%. These results use a lightweight patch-token MLP Encoder, indicating that the MSD can construct accurate future trajectories from comparatively shallow history representations. Overall, UVHF improves forecasting accuracy over the evaluated horizon-specific baselines while consolidating the four requested horizons into one model.

5.3. One-model-all-horizons evaluation

Section 5.2 shows that UVHF can outperform baselines trained separately for each of the four horizons. A stricter comparison places every method under the same one-model-all-horizons workflow. For each dataset, a single model produces the maximum-length forecast, and requests for $H \in \{96, 192, 336\}$ are evaluated from the corresponding output prefixes without retraining or reselection. This setting maintains consistency across overlapping predictions within each method and provides a more deployment-relevant comparison without horizon-specific model specialization.

Table 2 summarizes the four-horizon mean for each dataset, with complete horizon-wise results provided in Appendix B. Under the shared one-model constraint, UVHF ranks first in all 14 dataset–metric comparisons. Averaged over all seven datasets and four horizons, it reduces MSE and MAE by 6.45% and 3.72% relative to TimeAlign, the next strongest complete baseline in Table 2. These gains exceed the corresponding 4.94% and 2.54% improvements

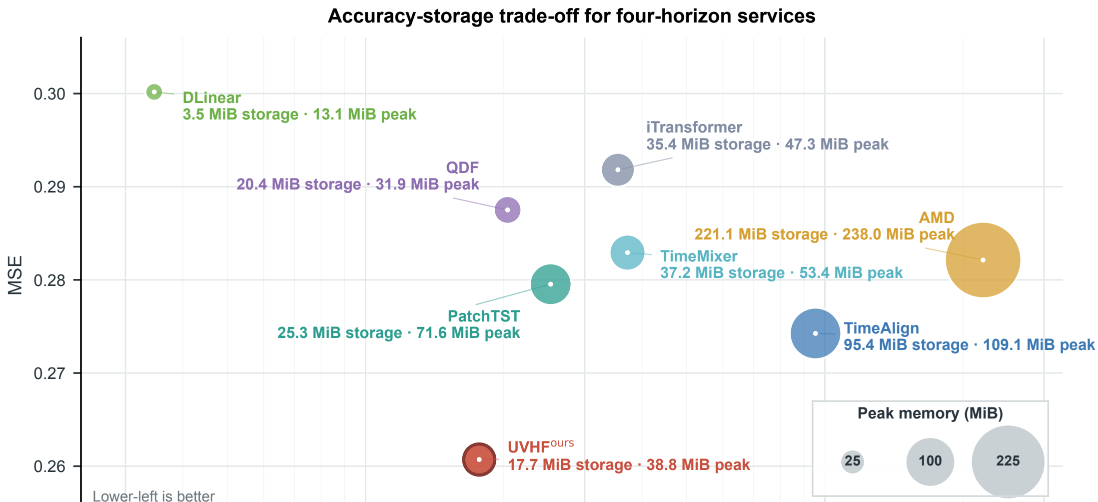

Figure 5: Accuracy and system cost for four-horizon forecasting. Macro MSE is plotted against total checkpoint storage, and bubble area represents peak allocated inference memory. The storage axis is logarithmic; lower-left positions indicate a more favorable trade-off.

in the horizon-specific comparison by 1.51 and 1.18 percentage points, respectively, showing that the advantage of UVHF becomes more pronounced when every method follows the same unified varied-horizon forecasting workflow.

These results establish UVHF as an effective framework for unified varied-horizon forecasting. Its MSD constructs one prefix-consistent trajectory from a scope-indexed forecast field, while Target-Adaptive Scope Allocation integrates predictive information across sharing granularities for each future step. The consistently strong results across the four evaluated horizons support this output-side design for forecasting requests ranging from short to long endpoints.

5.4. Accuracy and system cost

Forecast accuracy is only one consideration in practical forecasting services. Runtime memory and model storage also determine deployment and management costs, particularly in resource-constrained settings. We therefore measure the peak allocated GPU memory of the complete inference service and the total checkpoint storage required to serve all four horizons. Fig. 5 compares forecasting accuracy with these two costs.

As shown in Fig. 5, UVHF attains the lowest macro MSE among the eight displayed methods, at 0.261, while requiring 17.677 MiB of checkpoint storage and 38.817 MiB of peak allocated inference memory. Compared with TimeAlign, AMD, iTransformer, PatchTST and TimeMixer, the unified model reduces checkpoint storage by 30.12–92.01% and peak memory by 18.01–83.69%. Relative to TimeAlign in particular, UVHF lowers MSE by 4.94%, checkpoint storage by 81.48% and peak memory by 64.42%.

5.5. Component and training-objective ablations

To determine whether the individual UVHF components contribute to forecasting performance, we conduct strictly controlled ablations in which each variant is trained end to end under the same protocol. **w/o BCA** retains the Uniform-Prefix Forecasting Loss but removes the Scope-Wise Forecasting Loss and Allocation-Balance Regularizer. **w/o Target-Adaptive Allocation** replaces learned Scope Probabilities with equal non-adaptive fusion. **Shared Scope Projection** replaces the scope-specific history projections with one shared projection, whereas **Fixed Scope** ($s = 144$) retains only the preregistered middle scope.

As reported in Table 3, Full UVHF achieves a five-dataset macro MSE/MAE of 0.305/0.344 and ranks first in all 12 dataset and average metric columns. Removing BCA causes the largest aggregate decline, while equal fusion, Shared Scope Projection and Fixed Scope also increase both error metrics. Full UVHF improves MSE and MAE over every control on all five datasets.

Table 3: UVHF ablations (four-horizon mean MSE/MAE). Bold and underline indicate best and second-best values.

Variant	ETTm1		ETTm2		ETTh1		ETTh2		Weather		Avg	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Full UVHF	**0.342**	**0.365**	**0.253**	**0.309**	**0.407**	**0.432**	**0.311**	**0.367**	**0.214**	**0.245**	**0.305**	**0.344**
w/o BCA	0.353	0.378	<u>0.258</u>	0.315	0.434	0.450	0.319	0.375	0.218	0.250	0.316	0.354
w/o Target-Adaptive Allocation	<u>0.347</u>	<u>0.369</u>	0.259	0.317	<u>0.413</u>	0.437	<u>0.315</u>	<u>0.371</u>	<u>0.217</u>	<u>0.249</u>	<u>0.310</u>	<u>0.349</u>
Shared Scope Projection	0.355	0.376	<u>0.258</u>	<u>0.314</u>	0.415	<u>0.435</u>	0.325	0.382	<u>0.217</u>	0.250	0.314	0.351
Fixed Scope (s = 144)	0.351	0.372	0.263	0.319	0.428	0.441	0.318	0.373	<u>0.217</u>	<u>0.249</u>	0.315	0.351

The controlled comparisons support the utility of BCA, Target-Adaptive Scope Allocation, scope-specific projections and the multi-scope generation implemented by the MSD. Their joint design yields the strongest aggregate forecasting performance.

5.6. *Scope diversity and allocation behavior*

The ablation results support the contribution of the individual UVHF components to forecasting accuracy. We further examine whether the complete model exhibits the intended internal behavior. The analysis asks whether jointly trained scope lines retain distinct forecast signals and whether the Target-Adaptive Allocation Path produces diverse preferences across future regions. Region-specific forecast generation requires both properties: scope lines must avoid trajectory collapse, and allocation must vary across the future domain.

We concatenate the region-wise predictions from each scope line into its complete Scope-conditioned Forecast. Direct overlay obscures the differences among the five trajectories, so Fig. 6b visualizes their deviations from the fused trajectory, $\mathcal{F}_\tau^{(s)} - \widehat{y}_\tau$. The profiles exhibit distinct temporal structures, with a mean pairwise absolute disagreement of 0.0812 on the normalized scale. The scope-indexed forecast field therefore retains distinct candidate signals for region-wise integration and avoids collapse to repeated trajectories.

Fig. 6a shows how the Target-Adaptive Allocation Path combines these heterogeneous scope signals. The fused maximum-horizon forecast is coloured by the scope receiving the highest mean Scope Probability within each future region. All five scopes attain the highest weight in at least one of the eight regions, indicating that allocation avoids collapse to a fixed regional preference. Fig. 6c displays the complete per-step Scope Probabilities, whose relative ordering varies across future steps and regions. The farthest region assigns its highest mean weight to the largest sharing scope, $s = 720$, consistent with the role of broader information sharing in distal forecasting. The three panels illustrate how distinct scope signals and region-dependent soft allocation support region-specific forecast generation. These signals are fused into one shared trajectory, while prefix-based inference preserves CHPC with a maximum absolute CHPD of 0 across all 20 dataset–horizon checks.

5.7. *Generalization studies*

The MSD is designed as an Encoder-agnostic plug-in decoder that operates on Encoder-produced history representations. We examine this versatility using DLinear and PatchTST as representatives of lightweight linear and Transformer-based Encoders, respectively [1, 2]. For each backbone, we replace the Original Decoder with the MSD and train the resulting UVHF instantiation end to end using BCA.

As shown in Fig. 7, replacing the Original Decoder with the MSD and applying BCA improves forecasting performance on both example backbones. For DLinear, the replacement lowers MSE on all three datasets and reduces the macro average from 0.303 to 0.286, corresponding to an improvement of 5.61%. PatchTST shows smaller but consistent gains, with macro MSE decreasing from 0.282 to 0.276, corresponding to an improvement of 2.13%. Improvements with both a lightweight linear backbone and a Transformer-based backbone support the portability of the MSD and its role within UVHF.

5.8. *Visualization of forecasting performance and prefix consistency*

The aggregate comparisons assess forecasting accuracy across datasets and horizons; Fig. 8 complements them with a trajectory-level view of accuracy and CHPC. We compare UVHF with four independently trained horizon-specific TimeMixer models on a selected ETTm1 validation window for the LUFL variable. All forecasts are aligned

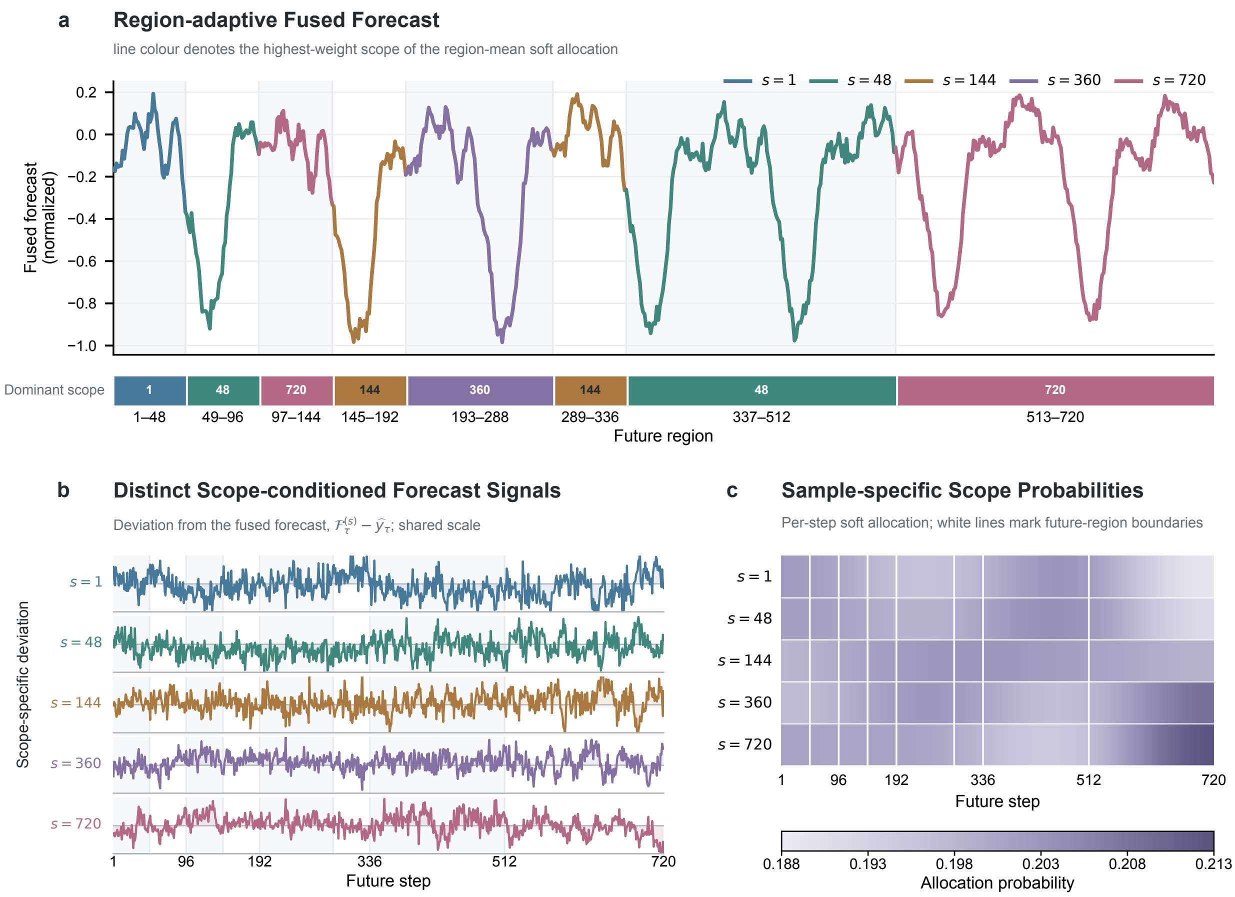

Figure 6: Scope diversity and sample-specific allocation behavior in UVHF. **a**, The fused forecast; line colour and the lower strip identify the highest-weight scope after averaging Scope Probabilities within each future region. **b**, Deviations of the five Scope-conditioned Forecasts from the fused forecast on a shared scale. **c**, Per-step Scope Probabilities; white lines mark future-region boundaries.

to the same forecast origin and future targets. Within the shared 96-step prefix, the TimeMixer models assign different values to the same future steps, whereas UVHF serves the four requested horizons through nested prefixes of one trajectory. Separate inference requests for UVHF returned identical predictions on their overlapping targets.

Across the full forecast window, UVHF more closely follows the recurring troughs, while the TimeMixer forecasts frequently overestimate these low-value intervals. The enlarged prefix makes both the differences among the TimeMixer predictions and UVHF's closer tracking of the observed trough directly visible. This selected case illustrates how forecasting accuracy and prefix consistency can coexist, while the aggregate comparisons in Sections 5.2 and 5.3 establish the broader performance results. Additional UVHF forecasting examples across all seven datasets are provided in Appendix C.

6. Discussion and Limitations

Unified varied-horizon forecasting. This task serves different request endpoints from a single prediction trajectory. Horizon-specific predictors optimize each endpoint independently and do not enforce CHPC across their overlapping outputs. UVHF maps a shared history representation and future-step coordinates to one trajectory, whose

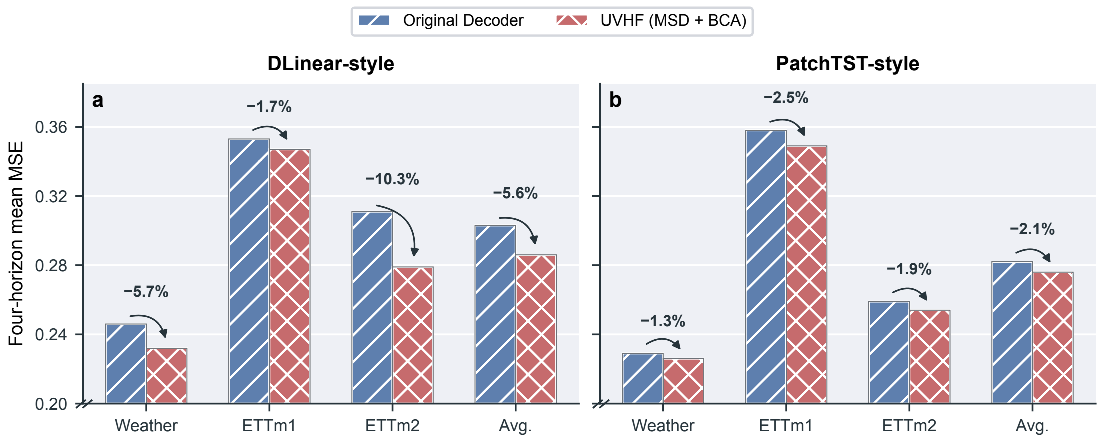

Figure 7: Generalization across two forecasting backbones. Four-horizon mean MSE is reported for **a**, DLinear and **b**, PatchTST with either the Original Decoder or UVHF (MSD trained with BCA).

prefixes answer requests of different lengths. The experiments in Sections 5.2 and 5.3 show that this structural consistency coexists with strong forecasting accuracy, and Fig. 5 demonstrates the deployment benefit of consolidating all horizon requests into one checkpoint.

Output-side multi-scope generation. Existing multi-scale encoders extract and combine historical dependencies at multiple input resolutions, where scale describes the granularity of history representation before decoding [23, 27, 28]. The MSD introduces a distinct output-side formulation in which candidate forecasts reuse history-conditioned latent states over different contiguous future extents. Each scope therefore specifies the sharing granularity of forecast generation, shifting multi-scale modeling from history representation to future construction. Evidence from the scope-wise ablations, the future-region analysis in Section 3 and the Fig. 6 diagnostics indicates that different future regions can benefit from different sharing extents. Target-Adaptive Allocation assigns step-specific preferences across these candidate forecast signals. Multi-scale encoders and the MSD address separate yet complementary stages of the forecasting pipeline. Multi-scale encoders organize historical evidence, while the MSD converts the resulting shared representation into forecasts at multiple future-sharing granularities.

Methodological limitations. The current formulation relies on a finite set of pre-specified contiguous scopes and future regions. This assumption may be restrictive when useful sharing patterns change at irregular locations, motivating data-adaptive region partitioning. Interactions among region forecasts are mediated through the shared History State and Future Coordinate. A dedicated cross-region forecasting mechanism could model dependencies that evolve across the future domain more explicitly. Multiple scope branches also increase decoder-side computation compared with a single-scope head, although shared encoding and unified serving reduce duplicated computation and storage. Future work should examine adaptive partitioning, cross-region interactions and more efficient multi-scope generation.

7. Conclusion

This work reframes multi-horizon service as the construction of one coherent future trajectory, with request endpoints exposed through nested prefixes. Within this formulation, CHPC defines the required cross-horizon coherence, and output-side sharing granularity becomes an explicit dimension of forecast generation. UVHF realizes this principle through an MSD that constructs Scope-conditioned Forecasts across multiple sharing extents and integrates them through Target-Adaptive Allocation. BCA supplies balanced supervision to the jointly optimized scope lines. The resulting framework combines region-sensitive forecast generation with prefix-consistent inference in a single horizon-agnostic model.

Across seven multivariate benchmarks, UVHF achieved state-of-the-art aggregate accuracy among the evaluated methods while consolidating every supported horizon into one model. Its advantage persisted under a common one-

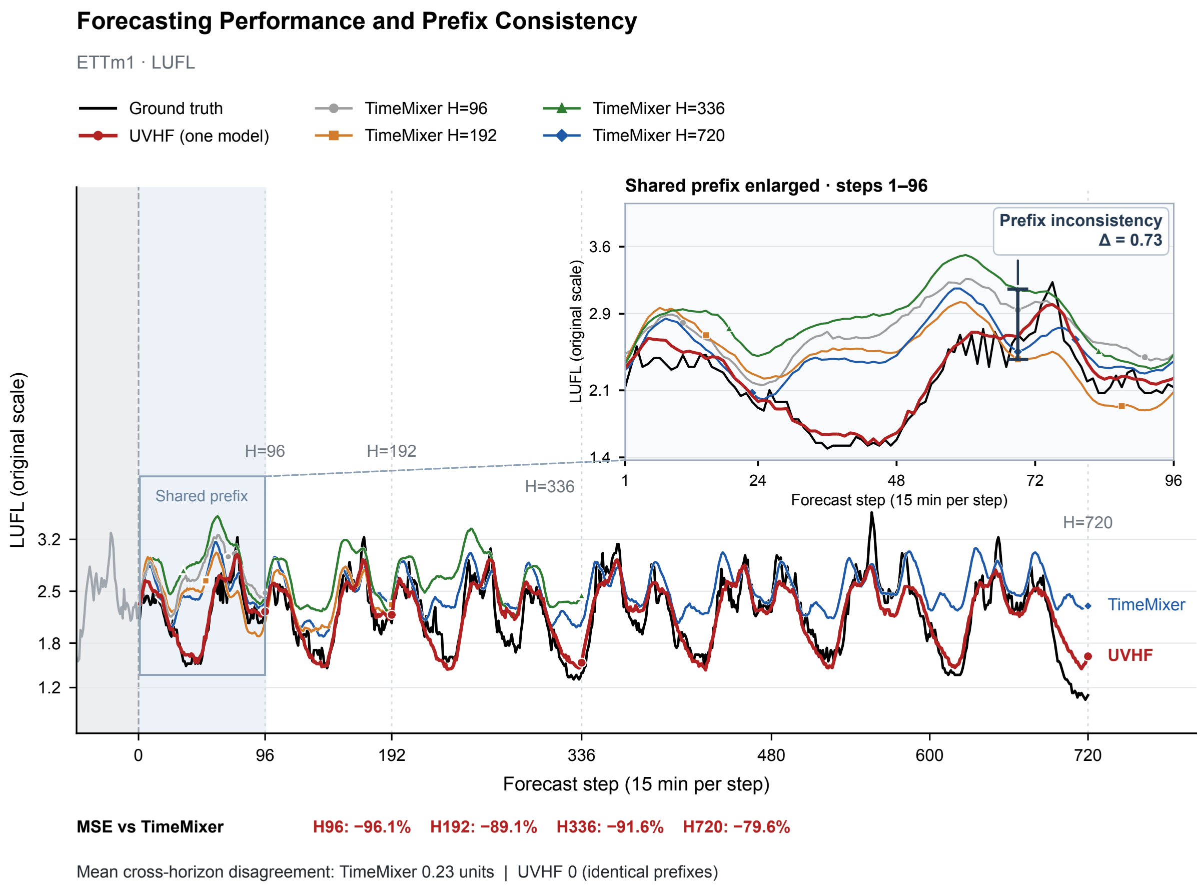

Figure 8: Forecasting accuracy and prefix consistency on a selected ETTm1–LUFL validation window. UVHF is compared with four horizon-specific TimeMixer models. The inset shows steps 1–96; Δ denotes their prediction range at step 69. MSE reductions use each complete horizon; mean disagreement averages the six pairwise mean absolute differences over their respective overlaps for LUFL.

model-all-horizons workflow, and controlled ablations and two-backbone studies supported component utility and MSD portability. These findings connect prefix-consistent serving with strong forecasting accuracy and efficient model consolidation in one framework. They also position output-side sharing granularity as a productive design axis for unified time-series forecasting.

CRediT authorship contribution statement

Chenhao Ying: Writing – original draft, Writing - review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation. **Jiangang Lu:** Writing - review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.1: Dataset statistics.

Dataset	Variables	Sampling Frequency	Dataset Size	Domain
ETTh1	7	1 h	(8545, 2881, 2881)	Electricity
ETTh2	7	1 h	(8545, 2881, 2881)	Electricity
ETTm1	7	15 min	(34465, 11521, 11521)	Electricity
ETTm2	7	15 min	(34465, 11521, 11521)	Electricity
ECL	321	1 h	(18317, 2633, 5261)	Electricity
Weather	21	10 min	(36792, 5271, 10540)	Weather
Solar	137	10 min	(36601, 5161, 10417)	Solar energy

Data availability

Data will be made available on request.

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Appendix A. Experiment Details

Appendix A.1. Metric Details

We use mean squared error (MSE) and mean absolute error (MAE) to evaluate forecasting accuracy. Given the ground-truth values y_i and their predictions $\widehat{y_i}$, the two metrics are defined as

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y_i})^2, \qquad \mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{y_i}\right|, \tag{A.1}$$

where N is the total number of scalar predictions over all evaluation windows, forecast steps and variables for the requested horizon. Lower MSE and MAE indicate more accurate forecasts.

Appendix A.2. Datasets

We evaluate UVHF on seven widely used multivariate time-series forecasting datasets spanning electricity transformers, electricity consumption, meteorology and solar generation. These benchmarks differ substantially in variable dimension, sampling frequency and temporal dynamics. Table A.1 summarizes their basic statistics.

1. **ETT (Electricity Transformer Temperature)** records seven load and temperature factors from two electricity transformers in two regions of China between 2016 and 2018 [10]. We use its four standard subsets: ETTh1 and ETTh2 are sampled hourly, whereas ETTm1 and ETTm2 are sampled every 15 minutes.
2. **Electricity (ECL)**[1] records the hourly electricity consumption of 321 clients from 2012 to 2014 and is provided by the UCI Machine Learning Repository.
3. **Weather**[2] is collected by the Beutenberg Weather Station at the Max Planck Institute for Biogeochemistry in Jena, Germany. It contains 21 meteorological indicators sampled every 10 minutes during 2020 [40].
4. **Solar** contains solar-power measurements collected every 10 minutes from 137 photovoltaic plants in Alabama during 2006 [41].

Dataset Size is reported as (Train, Validation, Test). The ETT datasets follow their standard fixed chronological splits, whereas Weather, ECL and Solar use chronological 70/10/20 splits. The scaler is fitted exclusively on the training segment. All datasets are evaluated at prediction lengths $H \in \{96, 192, 336, 720\}$.

[1] `https://archive.ics.uci.edu/dataset/321`
[2] `https://www.bgc-jena.mpg.de/wetter/`

Table A.2: Training and evaluation settings.

Dataset	Max Length	Initial LR	Batch	Grad. Accum.	Max epochs	Patience
ETTh1	720	3×10^{-4}	32	1	45	10
ETTh2	720	5×10^{-4}	32	1	30	7
ETTm1	720	1×10^{-4}	16	2	30	7
ETTm2	720	5×10^{-5}	16	2	60	12
Weather	720	2×10^{-5}	32	1	120	24
ECL	720	5×10^{-4}	4	8	30	7
Solar	720	3×10^{-4}	16	2	45	10

Table A.3: UVHF configuration.

Dataset	Scope set	Mode rank	Patch number	d_{model}	d_{ff}	Dropout	Weight decay	λ_{scope}	$\lambda_{\text{balance}}^{\max}/u_{\text{ramp}}$
ETTh1	$\{1, 48, 144, 360, 720\}$	109	24	32	32	0.1	0.01	1.0	0.1 / 0.25
ETTh2	$\{1, 48, 144, 360, 720\}$	116	12	64	128	0.1	0.01	1.0	0.1 / 0.25
ETTm1	$\{1, 48, 144, 360, 720\}$	116	1	128	256	0.9	0.01	1.0	0.1 / 0.25
ETTm2	$\{1, 48, 144, 360, 720\}$	64	6	128	128	0.2	0.001	1.0	0.1 / 0.25
Weather	$\{1, 48, 144, 360, 720\}$	116	19	64	128	0.0	0.01	1.0	0.1 / 0.25
ECL	$\{1, 48, 144, 360, 720\}$	64	1	256	1,024	0.3	0.01	1.0	0.1 / 0.25
Solar	$\{1, 48, 144, 360, 720\}$	128	4	256	256	0.3	0.01	1.0	0.1 / 0.25

Appendix A.3. Implementation Details

The local experiments are implemented in Python 3.12.13 and PyTorch 2.9.0 with CUDA 12.8. Each training run uses one NVIDIA GeForce RTX 3090 GPU. UVHF is trained from scratch with BCA using AdamW and a cosine learning-rate schedule [38, 39]. Table A.2 reports the dataset-specific optimization settings, and Table A.3 lists the corresponding Encoder interface and UVHF configuration.

Visualization protocol. The trajectory comparison in Section 5.8 uses separately reproduced TimeMixer models from the official codebase, with one model trained for each requested horizon and seed 2021. These checkpoints are distinct from the source of the TimeMixer results in Table 1. Each TimeMixer checkpoint was selected by its validation loss. The frozen UVHF checkpoint was queried separately at the four requested horizons to verify agreement on shared targets. The displayed LUFL window was selected after inspecting validation forecasts to make trajectory differences and prefix disagreement visible; it does not provide a representative estimate of the performance gap. Forecasts were aligned by origin and target timestamp and transformed back to the original data scale using training-set statistics.

Gradient Accumulation Steps denotes the number of mini-batches whose gradients are accumulated before one optimizer update; the effective batch size is therefore the batch size multiplied by this value. All runs use seed 2021. The checkpoint with the lowest validation mean MSE over $H \in \{96, 192, 336, 720\}$ is retained and serves all four forecasting requests. Weight decay is 0.01 except for ETTm2, where it is 0.001, and early stopping uses zero minimum improvement.

The scope set and BCA coefficients are shared across datasets. Specifically,

$$\lambda_{\text{scope}} = 1, \qquad \lambda_{\text{balance}}(u) = \lambda_{\text{balance}}^{\max} \min\Big(u/u_{\text{ramp}}, 1\Big), \tag{A.2}$$

where $u \in [0, 1]$ denotes normalized optimizer progress, $\lambda_{\text{balance}}^{\max}$ is the maximum balance weight and u_{ramp} is the progress at which this weight is reached. We use $\lambda_{\text{balance}}^{\max} = 0.1$ and $u_{\text{ramp}} = 0.25$ for every dataset.

Appendix B. Full Results

Tables B.1 and B.2 report the complete horizon-wise MSE and MAE results underlying Sections 5.2 and 5.3. Table B.1 compares UVHF with horizon-specific baselines, while Table B.2 evaluates every method under the one-model-all-horizons workflow. The main text reports the corresponding four-horizon dataset averages.

Table B.1: Horizon-wise results against horizon-specific baselines. Bold and underline indicate best and second-best values. Best and Second-best count the corresponding ranks across 28 dataset–horizon pairs.

Dataset	H	UVHF (Ours)		TimeAlign (2026)		QDF (2026)		AMD (2025)		SimpleTM (2025)		TVNet (2025)		iTransformer (2024)		TimeMixer (2024)		ModernTCN (2024)		PatchTST (2023)		DLinear (2023)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	**0.265**	**0.326**	0.280	0.331	0.283	0.336	0.289	0.343	0.325	0.363	0.288	0.343	0.300	0.353	0.293	0.345	0.292	0.346	0.293	0.346	0.300	0.345
	192	**0.304**	**0.349**	0.323	0.357	0.336	0.369	0.329	0.366	0.361	0.381	0.326	0.367	0.345	0.382	0.335	0.372	0.332	0.368	0.333	0.370	0.336	0.366
	336	**0.343**	0.371	0.352	0.376	0.369	0.390	0.365	0.386	0.391	0.404	0.365	0.391	0.374	0.398	0.368	0.386	0.365	0.391	0.369	**0.369**	0.367	0.386
	720	**0.404**	**0.405**	0.408	0.409	0.423	0.424	0.423	0.417	0.454	0.438	0.412	0.413	0.429	0.430	0.426	0.417	0.416	0.417	0.416	0.420	0.419	0.416
	Avg.	**0.329**	**0.363**	0.341	0.368	0.353	0.380	0.351	0.378	0.383	0.396	0.348	0.379	0.362	0.391	0.356	0.380	0.351	0.381	0.353	0.376	0.356	0.378
ETTm2	96	0.159	**0.242**	**0.157**	0.244	0.164	0.253	0.168	0.258	0.175	0.258	0.161	0.254	0.175	0.266	0.165	0.256	0.166	0.256	0.166	0.256	0.164	0.255
	192	0.218	0.285	**0.210**	**0.280**	0.218	0.291	0.221	0.295	0.240	0.301	0.220	0.293	0.242	0.312	0.225	0.298	0.222	0.293	0.223	0.296	0.224	0.304
	336	0.270	0.318	**0.265**	0.318	0.282	0.330	0.270	0.327	0.305	0.342	0.272	**0.316**	0.282	0.340	0.277	0.332	0.272	0.324	0.274	0.329	0.277	0.337
	720	**0.343**	**0.371**	0.346	0.374	0.364	0.385	0.355	0.381	0.401	0.399	0.349	0.379	0.378	0.398	0.360	0.387	0.351	0.381	0.362	0.385	0.371	0.401
	Avg.	0.248	**0.304**	**0.245**	**0.304**	0.257	0.315	0.254	0.315	0.280	0.325	0.251	0.311	0.269	0.329	0.257	0.318	0.253	0.314	0.256	0.317	0.259	0.324
ETTh1	96	**0.348**	**0.386**	0.380	0.407	0.370	0.394	0.371	0.399	0.367	0.393	0.371	0.408	0.386	0.405	0.372	0.401	0.368	0.394	0.377	0.397	0.379	0.403
	192	**0.377**	**0.403**	0.406	0.419	0.422	0.427	0.403	0.420	0.429	0.427	0.398	0.409	0.424	0.440	0.413	0.430	0.405	0.413	0.409	0.428	0.408	0.419
	336	0.393	0.414	0.427	0.432	0.464	0.455	0.423	0.432	0.447	0.445	0.401	**0.409**	0.449	0.460	0.438	0.450	**0.391**	0.412	0.431	0.444	0.440	0.440
	720	**0.443**	**0.458**	0.459	0.464	0.499	0.493	0.452	0.461	0.470	0.465	0.458	0.459	0.495	0.487	0.486	0.484	0.450	0.461	0.457	0.477	0.471	0.493
	Avg.	**0.390**	**0.415**	0.418	0.431	0.439	0.442	0.412	0.428	0.428	0.433	0.407	0.421	0.439	0.448	0.427	0.441	0.404	0.420	0.419	0.437	0.425	0.439
ETTh2	96	**0.241**	**0.314**	0.270	0.331	0.283	0.342	0.279	0.343	0.285	0.342	0.263	0.329	0.297	0.348	0.281	0.351	0.263	0.332	0.274	0.337	0.289	0.353
	192	**0.282**	**0.345**	0.334	0.374	0.359	0.391	0.363	0.397	0.357	0.387	0.319	0.372	0.371	0.403	0.349	0.387	0.320	0.374	0.348	0.384	0.383	0.418
	336	**0.310**	**0.369**	0.376	0.405	0.361	0.401	0.381	0.419	0.369	0.402	0.311	0.373	0.404	0.428	0.366	0.413	0.313	0.376	0.377	0.416	0.448	0.465
	720	**0.392**	**0.429**	0.406	0.436	0.407	0.440	0.442	0.467	0.414	0.436	0.401	0.434	0.424	0.444	0.401	0.436	**0.392**	0.433	0.406	0.441	0.605	0.551
	Avg.	**0.306**	**0.364**	0.347	0.387	0.353	0.394	0.366	0.406	0.356	0.392	0.324	0.377	0.374	0.406	0.349	0.397	0.322	0.379	0.351	0.395	0.431	0.447
Weather	96	**0.140**	**0.180**	0.143	0.182	0.152	0.202	0.148	0.203	0.157	0.204	0.147	0.198	0.159	0.208	0.147	0.198	0.149	0.200	0.149	0.198	0.170	0.230
	192	**0.181**	**0.223**	0.185	0.224	0.205	0.251	0.193	0.243	0.205	0.248	0.194	0.238	0.200	0.248	0.192	0.243	0.196	0.245	0.194	0.241	0.216	0.273
	336	**0.231**	**0.263**	0.235	**0.263**	0.248	0.282	0.242	0.281	0.264	0.291	0.235	0.277	0.253	0.289	0.247	0.284	0.238	0.277	0.245	0.282	0.258	0.307
	720	**0.302**	**0.313**	0.308	0.319	0.341	0.351	0.315	0.332	0.346	0.344	0.308	0.331	0.321	0.338	0.318	0.330	0.314	0.334	0.314	0.334	0.323	0.362
	Avg.	**0.214**	**0.245**	0.218	0.247	0.237	0.271	0.225	0.265	0.243	0.271	0.221	0.261	0.233	0.271	0.226	0.264	0.224	0.264	0.226	0.264	0.242	0.293
ECL	96	**0.116**	**0.213**	0.128	0.218	0.127	0.222	0.131	0.228	0.143	0.237	0.142	0.223	0.138	0.237	0.142	0.247	0.129	0.226	0.129	0.222	0.140	0.237
	192	**0.136**	**0.231**	0.145	0.235	0.151	0.244	0.151	0.244	0.151	0.248	0.165	0.241	0.157	0.256	0.168	0.269	0.143	0.239	0.147	0.240	0.152	0.249
	336	**0.157**	**0.251**	0.162	**0.251**	0.167	0.260	0.167	0.262	0.176	0.269	0.164	0.269	0.167	0.264	0.171	0.260	0.161	0.259	0.163	0.259	0.170	0.267
	720	0.195	0.280	**0.190**	**0.278**	0.223	0.314	0.200	0.292	0.201	0.293	**0.190**	0.284	0.194	0.286	0.209	0.313	0.191	0.286	0.197	0.290	0.203	0.301
	Avg.	**0.151**	**0.244**	0.156	0.246	0.167	0.260	0.162	0.257	0.168	0.262	0.165	0.254	0.164	0.261	0.173	0.272	0.156	0.253	0.159	0.253	0.166	0.264
Solar	96	**0.166**	**0.195**	0.182	0.205	0.193	0.232	0.188	0.235	0.169	0.229	0.204	0.260	0.188	0.232	0.179	0.232	0.202	0.263	0.178	0.229	0.197	0.210
	192	**0.185**	**0.206**	0.198	0.217	0.207	0.261	0.203	0.246	0.189	0.252	0.227	0.272	0.193	0.248	0.201	0.259	0.223	0.279	0.189	0.246	0.218	0.222
	336	0.195	**0.213**	0.203	0.222	0.210	0.263	0.213	0.252	0.199	0.262	0.241	0.288	0.203	0.249	**0.190**	0.256	0.241	0.292	0.198	0.249	0.234	0.231
	720	0.204	**0.218**	**0.201**	0.223	0.223	0.278	0.215	0.257	0.207	0.259	0.242	0.287	0.223	0.261	0.203	0.261	0.247	0.292	0.209	0.256	0.243	0.241
	Avg.	**0.188**	**0.208**	0.196	0.217	0.208	0.258	0.205	0.248	0.191	0.251	0.229	0.277	0.202	0.248	0.193	0.252	0.228	0.282	0.194	0.245	0.223	0.226
Best		21	23	5	4	0	0	0	0	0	0	1	2	0	0	1	0	2	0	0	1	0	0
Second-best		5	4	9	15	2	0	1	0	3	1	7	6	0	0	2	0	5	4	1	1	0	0

Table B.2: Horizon-wise results under the one-model-all-horizons protocol. Bold and underline indicate best and second-best values. Best and Second-best count the corresponding ranks across 28 dataset–horizon pairs.

Dataset	H	UVHF (Ours)		TimeAlign (2026)		QDF (2026)		AMD (2025)		SimpleTM (2025)		iTransformer (2024b)		PatchTST (2023)		DLinear (2023)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	**0.265**	**0.326**	0.290	0.338	0.292	0.348	0.294	0.348	0.326	0.368	0.346	0.382	0.298	0.350	0.301	0.346
	192	**0.304**	**0.349**	0.325	0.358	0.330	0.371	0.333	0.368	0.362	0.385	0.384	0.399	0.335	0.372	0.336	0.367
	336	**0.343**	**0.371**	0.355	0.378	0.364	0.391	0.366	0.387	0.392	0.406	0.418	0.420	0.367	0.392	0.371	0.387
	720	**0.404**	**0.405**	0.408	0.409	0.423	0.424	0.423	0.417	0.454	0.438	0.487	0.456	0.415	0.421	0.427	0.423
	Avg.	**0.329**	**0.363**	0.345	0.371	0.352	0.384	0.354	0.380	0.384	0.399	0.409	0.415	0.354	0.384	0.359	0.381
ETTm2	96	**0.159**	**0.242**	0.168	0.253	0.165	0.255	0.170	0.260	0.186	0.270	0.190	0.277	0.169	0.258	0.189	0.289
	192	0.218	**0.285**	**0.216**	0.286	0.223	0.294	0.222	0.295	0.247	0.308	0.253	0.315	0.223	0.295	0.256	0.339
	336	0.270	**0.318**	**0.268**	0.320	0.276	0.330	0.271	0.327	0.305	0.344	0.318	0.354	0.275	0.329	0.320	0.381
	720	**0.343**	**0.371**	0.346	0.374	0.364	0.385	0.355	0.381	0.401	0.399	0.416	0.409	0.365	0.383	0.403	0.428
	Avg.	**0.248**	**0.304**	0.250	0.308	0.257	0.316	0.255	0.316	0.285	0.330	0.294	0.339	0.258	0.316	0.292	0.359
ETTh1	96	**0.348**	**0.386**	0.375	0.404	0.401	0.418	0.378	0.403	0.382	0.406	0.409	0.422	0.394	0.414	0.416	0.436
	192	**0.377**	**0.403**	0.411	0.425	0.441	0.441	0.410	0.424	0.434	0.431	0.457	0.448	0.429	0.433	0.457	0.465
	336	**0.393**	**0.414**	0.433	0.437	0.467	0.459	0.426	0.433	0.448	0.442	0.496	0.467	0.440	0.442	0.484	0.481
	720	**0.443**	**0.458**	0.459	0.464	0.499	0.493	0.452	0.461	0.470	0.465	0.509	0.494	0.455	0.469	0.538	0.538
	Avg.	**0.390**	**0.415**	0.420	0.433	0.452	0.453	0.416	0.430	0.434	0.436	0.468	0.458	0.429	0.440	0.474	0.480
ETTh2	96	**0.241**	**0.314**	0.296	0.349	0.272	0.340	0.285	0.351	0.285	0.344	0.303	0.353	0.277	0.339	0.322	0.382
	192	**0.282**	**0.345**	0.364	0.393	0.334	0.382	0.374	0.405	0.358	0.388	0.385	0.401	0.345	0.382	0.367	0.408
	336	**0.310**	**0.369**	0.405	0.418	0.358	0.404	0.379	0.421	0.367	0.404	0.424	0.433	0.379	0.414	0.385	0.425
	720	**0.392**	**0.429**	0.406	0.436	0.407	0.440	0.442	0.467	0.414	0.436	0.430	0.446	0.407	0.443	0.584	0.543
	Avg.	**0.306**	**0.364**	0.368	0.399	0.343	0.392	0.370	0.411	0.356	0.393	0.385	0.408	0.352	0.395	0.415	0.439
Weather	96	**0.140**	**0.180**	0.148	0.187	0.191	0.246	0.149	0.204	0.177	0.222	0.179	0.222	0.155	0.205	0.177	0.240
	192	**0.181**	**0.223**	0.187	0.226	0.231	0.276	0.194	0.244	0.221	0.259	0.228	0.262	0.199	0.246	0.218	0.278
	336	**0.231**	**0.263**	0.238	0.268	0.276	0.307	0.244	0.282	0.273	0.296	0.283	0.301	0.249	0.284	0.264	0.317
	720	**0.302**	**0.313**	0.308	0.319	0.341	0.351	0.315	0.332	0.346	0.344	0.359	0.351	0.319	0.335	0.328	0.369
	Avg.	**0.214**	**0.245**	0.220	0.250	0.260	0.295	0.226	0.266	0.254	0.280	0.262	0.284	0.231	0.268	0.247	0.301
ECL	96	**0.116**	**0.213**	0.129	0.220	0.155	0.258	0.132	0.230	0.150	0.246	0.152	0.245	0.132	0.228	0.140	0.238
	192	**0.136**	**0.231**	0.145	0.235	0.169	0.269	0.152	0.246	0.152	0.251	0.164	0.257	0.148	0.243	0.154	0.251
	336	**0.157**	**0.251**	0.163	0.252	0.185	0.284	0.166	0.261	0.173	0.269	0.179	0.273	0.164	0.259	0.169	0.268
	720	0.195	0.280	**0.190**	**0.278**	0.223	0.314	0.200	0.292	0.201	0.293	0.211	0.301	0.201	0.291	0.204	0.301
	Avg.	**0.151**	**0.244**	0.157	0.246	0.183	0.281	0.163	0.257	0.169	0.265	0.177	0.269	0.161	0.255	0.167	0.265
Solar	96	**0.166**	**0.195**	0.179	0.211	0.186	0.256	0.190	0.242	0.171	0.233	0.197	0.239	0.181	0.235	0.222	0.292
	192	**0.185**	**0.206**	0.192	0.218	0.205	0.267	0.202	0.249	0.187	0.246	0.228	0.259	0.190	0.245	0.250	0.309
	336	0.195	**0.213**	0.198	0.223	0.215	0.272	0.210	0.254	**0.193**	0.251	0.243	0.271	0.198	0.251	0.269	0.324
	720	0.204	**0.218**	**0.201**	0.223	0.223	0.278	0.215	0.257	0.207	0.259	0.250	0.276	0.206	0.257	0.271	0.327
	Avg.	**0.188**	**0.208**	0.193	0.219	0.207	0.268	0.204	0.250	0.190	0.247	0.229	0.261	0.194	0.247	0.253	0.313
Best		23	27	4	1	0	0	0	0	1	0	0	0	0	0	0	0
Second-best		5	1	14	20	4	2	3	4	2	2	0	0	0	2	0	0

Appendix C. Visualization

Complementing the baseline comparison in Section 5.8, Fig. C.1 presents additional UVHF forecasting examples for the seven paper-core datasets. For each dataset, two validation samples show the ground-truth future and the fused UVHF prediction over the maximum target length $T = 720$. The paired rulers and vertical guides mark the four supported prediction lengths, showing how the same predicted trajectory provides the prefixes requested at $H \in \{96, 192, 336, 720\}$.

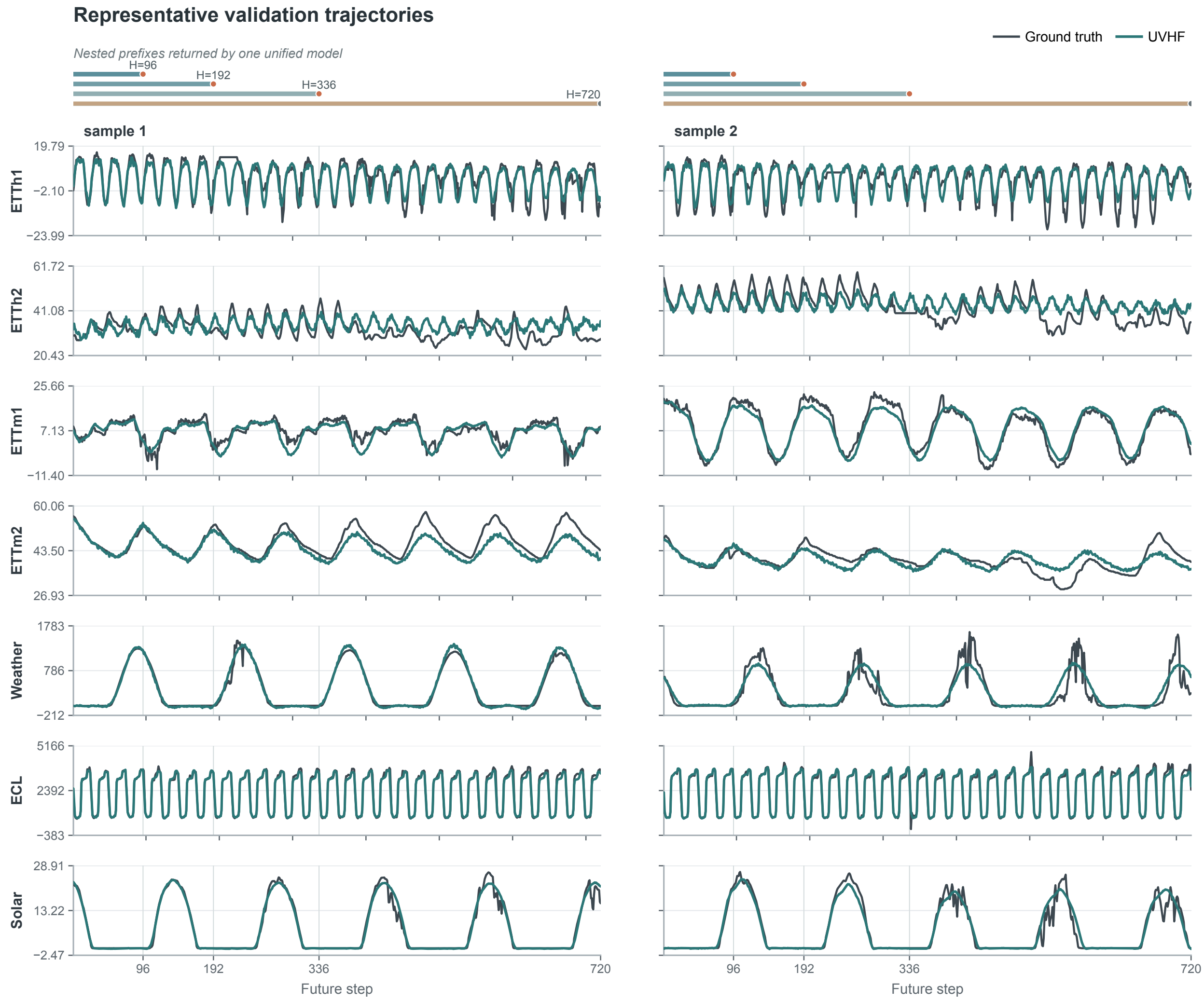

Figure C.1: Visualization of varied-horizon forecasts across seven datasets. The dark curve denotes the ground truth and the teal curve denotes the prediction from UVHF. Two validation examples are shown for each dataset. The paired rulers and vertical guides identify the four supported prediction lengths.

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